Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/18/2018
PI Name:	Venkateswaran, Kasthuri Ph.D.		
Project Title:	ISS Microbial Observatory - a Genetic Approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
PI Email:	kjvenkat@jpl.nasa.gov	**Fax:**	FY 818-393-4176
PI Organization Type:	NASA CENTER	**Phone:**	818-393-1481
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	Biotechnology and Planetary Protection		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	09/30/2018
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS Microbial Observatory NOTE: Extended to 9/30/2018 per F. Hernandez/ARC (Ed., 3/26/18) NOTE: Extended to 12/31/2017 per F. Hernandez/ARC (Ed., 8/22/17) NOTE: Extended to 04/30/2017 per F. Hernandez/ARC (Ed., 2/2/17) NOTE: Extended to 12/31/2016 per F. Hernandez/ARC (Ed., 4/11/16) NOTE: Extended to 4/30/2016 per A. Chu/ARC (Ed., 9/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fox, George Ph.D. (University of Houston) Pierson, Duane Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	19-12829-26 under Task Order NNN13D111T		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	As recommended by the National Research Council Decadal Survey, the Jet Propulsion Laboratory (JPL) proposes to establish an International Space Station-Microbial Observatory (ISS-MO) to generate a microbial census of the space station's surfaces and atmosphere using advanced molecular microbial community analysis techniques, supported by traditional culture-based methods and modern bioinformatic computational modeling. The proposed ISS-MO establishment will lead to long-term, multigenerational studies of microbial population dynamics, addressing recommendation P1 of this solicitation. It will also provide answers to NASA Space Biology Guiding Questions CMM-1, CMM-4, and CMM-5. The ISS-MO project's methodology will serve as the foundation for an extensive microbial census, offering significant insight into spaceflight-induced changes in the populations of beneficial and potentially harmful microbes. Additionally, it will provide NASA with both a mechanistic understanding of these changes (e.g., cataloging population changes and mapping/linking these to environmental niche and genomic changes), as well as insight into practical countermeasures for mitigating risks to humans and environmental systems. The ISS-MO team will use existing ISS sample collection technologies to generate an initial microbial census. Following their return to Earth, samples from the various ISS modules will be analyzed using standardized technologies from the Mars Program-funded projects. The proposed ISS-MO project includes the delivery of a database that compiles the genomic sequences and genetic information for all the microbes encountered within the ISS habitat. Using this data, NASA can more accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to providing microbial profiles, the ISS-MO team will identify which microbial taxa pose particular threats to crew health. Furthermore, specific ISS-MO project aims will enable us to resolve applicable NASA-Human Research Program integrated research plan risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NASA crewmembers are exposed constantly to the dust in a closed system, and have also reported allergies. Subsequent microbiological analyses using traditional methods noticed no anomaly and were also not able to predict the causative microbial agents. The next-generation sequencing study provided NASA with the ability to assess the spectrum of microorganisms associated with the ISS surfaces and would help to predict possible microbial agents that might affect crew health. The results of the present study also exhibited presence of representative human associated (skin, gastrointestinal track, wounds, etc.) bacterial commensals and opportunistic fungal pathogens. Furthermore, based on the phylogenetic affiliation it is possible to determine whether these viable microorganisms are hardy and robust that have been previously reported to survive in oligotrophic environments (cleanrooms, hospital, commercial airliner, etc.). The study has improved our understanding of background microbial contamination, thus facilitating the development of biosensors to monitor closed habitats like ISS and future manned missions.
Task Progress:	Objectives. The International Space Station (ISS) is a closed system inhabited by microorganisms originating from life support systems, cargo and crew, that are exposed to unique selective pressures such as microgravity. The NASA Space Biology program funded a study to catalogue ISS environmental microbiome (Microbial Tracking – 1; MT-1). Several defined locations were systematically collected from 3 consecutive flights and microbial census was generated using advanced molecular microbial community analysis “omics” techniques, supported by traditional culture-based methods. Similarly, an Inflated Lunar/Mars Analog Habitat as a ground control system and SpaceX CRS (Commercial Resupply Service) vehicles to trace source of any possible contamination were examined and results presented. The 3-year funded 5 year time period of the MT-1 experiments was completed and the information produced would serve as the foundation for an extensive microbial census, offering significant insight into spaceflight-induced changes in the populations of beneficial and potentially harmful microbes. The safety of crewmembers and the maintenance of hardware are the primary goals for monitoring microorganisms in this closed habitat. This study provides the first comprehensive catalogue of both total and intact/viable bacteria and fungi found on surfaces in closed space systems and can be used to help develop safety measures that meet NASA requirements for deep space human habitation. The results of this study can have significant impact on our understanding of other confined built environments on the Earth such as clean rooms used in the pharmaceutical and medical industries. The outcomes of the MT-1 microbiome study presented here may enable the development of suitable countermeasure technologies allowing to eradicate problematic microorganisms for human habitation in space and in confined built environments on Earth (e.g., medical, pharmaceutical, and spacecraft cleanrooms). These data sets were placed in the NASA GeneLab that is open to the scientific research community to encourage innovation.
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